

July 27, 2026 09:00 PM GMT

Greater China Semiconductors | Asia Pacific

AI Networking: Scale-up Technology Enables China's AI Supernodes

Following our [WAIC takeaways](#), we further examine China's emerging scale-up landscape, the role of optical interconnect, and implications for the domestic interconnect supply chain.

AI Supernodes enabled by scale-up are becoming the focus of China's AI computing landscape. Alongside our China [WAIC takeaways](#), we provide an in-depth follow-up study of China's scale-up technology development. As we wrote in [China's AI Accelerators – Who's Poised to Win?](#), Chinese AI GPUs are strong in optical networking and server rack design at the system level, although they remain constrained by wafer process technology at the chip level ([Exhibit 6](#)). These chip-level constraints are shifting competition in China's AI computing market from standalone chip specs to system solutions. WAIC 2026 featured fewer chip launches and more supernode solutions. Leading domestic accelerator vendors displayed rack-scale or multi-rack supernodes with scale-up domains of 64 accelerators or more, mostly enabled by proprietary scale-up technologies.

China's scale-up landscape is rapidly broadening, led by proprietary fabrics but spanning multiple system architectures ([Exhibit 4](#)). A year ago, our US semis team published [Semiconductors: Scaling the AI Opportunity: A Scale-Up Network Primer \(29 Aug 2025\)](#); China is now catching up quickly in scale-up technology. Huawei remains the domestic scale leader, with Atlas 950 using UnifiedBus 2.0 to connect 1,024 NPUs in the demonstrated configuration and designed to scale to 8,192 NPUs. Biren's next-gen BR2xx NPO architecture targets up to 1,024 GPUs through BLink 2.0. Sugon's scaleX640 connects 640 accelerators in a single rack. Moore Threads(not covered) MTT C256 supports 128 GPUs in one rack and 256 across two racks through MT-Link 2.0, while MetaX, Alibaba and Enflame support 64- to 128-accelerator domains through MetaXLink-E, ICN, and GCU-LARE, respectively.

Stock implications: We believe **Montage** (OW, covered by Daniel Yen) is set to benefit from higher PCIe interconnect content in larger, distributed supernodes. As systems scale across racks, denser and longer PCIe links among CPUs, xPUs, switches, NICs and peripherals should support demand for PCIe retimers and PCIe switch chips. We also expect China AI GPU vendors, such as **Hygon** (OW, covered by Daisy Dai) (along with its ecosystem partner Sugon), **Cambricon** and **Iluvatar** (both OW, covered by Charlie Chan), to introduce or leverage scale-up technology to improve their AI server rack performance.

(Continued below.)

MORGAN STANLEY ASIA LIMITED+

Ethan Jia

Research Associate

Ethan.Jia@morganstanley.com

+852 3963-2287

MORGAN STANLEY TAIWAN LIMITED+

Charlie Chan

Equity Analyst

Charlie.Chan@morganstanley.com

+886 2 2730-1725

Daniel Yen, CFA

Equity Analyst

Daniel.Yen@morganstanley.com

+886 2 2730-2863

MORGAN STANLEY ASIA LIMITED+

Daisy Dai, CFA

Equity Analyst

Daisy.Dai@morganstanley.com

+852 2848-7310

Henry Zhao

Research Associate

Henry.Zhao@morganstanley.com

+852 2239-7731

MORGAN STANLEY TAIWAN LIMITED+

Tiffany Yeh

Equity Analyst

Tiffany.Yeh@morganstanley.com

+886 2 7712-3032

Lucas Wang

Research Associate

Lucas.Wang@morganstanley.com

+886 2 2730-2875

**GREATER CHINA TECHNOLOGY SEMICONDUCTORS**

Asia Pacific

Industry View

Attractive

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

China AI Interconnect - Scale-up

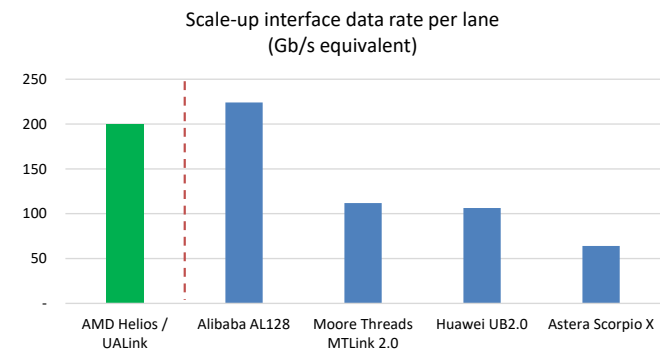
Global AI server scale-up technology trend

What is scale-up networking?

Scale-up networking refers to high-speed communication between GPUs/accelerators within the same server or rack. Interconnecting accelerators allow them to function as one large supercomputer, accessing the same memory and processing the same workload. This differs from scale-out networking, which connects multiple systems across a broader data center infrastructure.

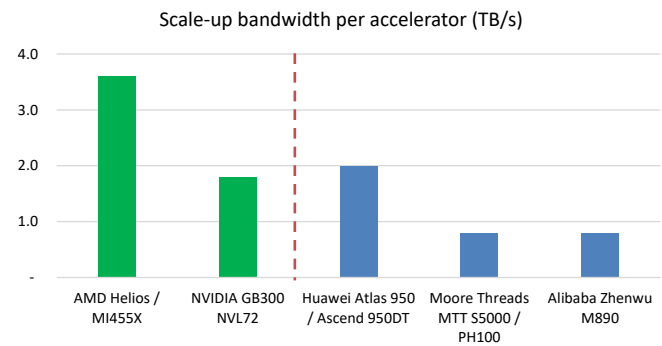
Montage's PCIe 6.x/CXL3.x x16 AEC has completed interoperability testing, and targets supernode and inter-rack deployments. Chinese systems deploying more xPUs per unit of compute could further increase PCIe content. Beyond standardized PCIe connectivity, multi-rack scale-up should also accelerate demand for domestic optical interconnect.

Exhibit 1: Scale-up interface data rate per lane



Source: Company data, Morgan Stanley Research

Exhibit 2: Scale-up bandwidth per accelerator



Source: Company data, Morgan Stanley Research

Exhibit 3: US AI networking scale-up technologies

Technology	NVLink5	UALink 1.0	SUE	PCIe Gen 6	ICI	NeruonLink
Purpose	GPU-GPU; CPU-GPU	xPU-xPU	xPU-xPU	Primarily CPU-XPU; xPU-xPU	TPU interconnect	Trainium interconnect
Bandwidth	100 GB/s	100 GB/s	100 GB/s	64 GT/s	50-100 c B/S	not standardized
Latency	10-20 nm	<100 nm	-	-	varies	varies
Ecosystem	NVIDIA (proprietary)	Open standard	Open standard	Open standard	Google (proprietary)	Amazon (proprietary)
Scalability	576 GPUs	1,024 xPUs	1,024 xPUs	varies	varies	64 Trainium

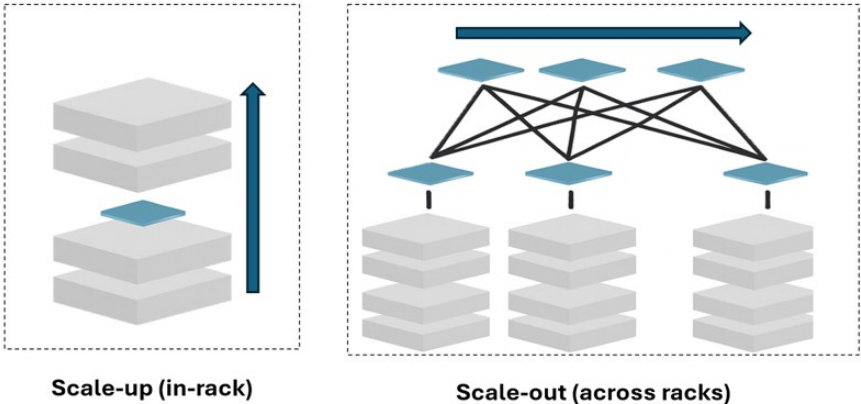
Source: Company data, Morgan Stanley Research

Exhibit 4: AI supernode scale-up solutions

Scale-up tech summary	NVIDIA	AMD	Asteralabs	Huawei	Moore Threads	Sugon	MetaX	Alibaba	Enflame	Enflame	Biren	Biren
Supernode / accelerator name	GB300 NVL72 (144 dies)	AMD Helios / MI455X	N/A	Atlas 950 SuperPoD	MTT C256 / MTT S5000, PH100	scaleX640	Xijing S600 / C600	Panjiu AL128 / Zhenwu M890	Yunsui ES164-O	NPO optical prototype	LightSphere 128 / Bili 1661	BR2x NPO supernode (announced)
Scale-up solution	NVLink 5 + NVSwitch	UALink 1.0 / UALoE	Scorpio X-Series	UnifiedBus 2.0	MTLink 2.0	scaleX fabric	MetaLink and MetaXLink-E	ALink system; M890 implements ICN + ICN Switch 1.0	GCU-LARE	Undisclosed	Undisclosed	Blink 2.0
Technology foundation	Proprietary	Ethernet PHY-based, memory-semantic UALink fabric	PCIe Gen6	Proprietary	Proprietary	Open multi-vendor system architecture	Proprietary	Supports proprietary protocol and UALink	Proprietary	Undisclosed	SiPh OCS	Proprietary
Singal lane speed	Physical-lane rate N/A; 100GB/s per NVLink 5 link	200GB/s per lane; 212.5GT/s raw signaling rate	Undisclosed	106.25Gb/s PAM4 under UBR2.0 specification	112Gb/s per lane	Undisclosed	Undisclosed	112G/234G SerDes supported at AL128 platform level	Undisclosed	Undisclosed	Undisclosed	Undisclosed
Total bandwidth per accelerator	1.8TB/s	3.6TB/s	Undisclosed	2TB/s	800GB/s	Undisclosed	Undisclosed	800GB/s	Undisclosed	Undisclosed	Undisclosed	Undisclosed
Total accelerator units in supernode	72	72	Up to 80 per switch (merchant solution)	1,024 NPUs demonstrated / 8,192 NPUs announced	128 GPUs in one rack; 256 GPUs across two racks	640	64 GPUs in 5600 rack; 128 GPUs through C600 MetaXLink-E	128 per rack; 64 per disclosed full-bandwidth ICN domain	64	512-plus	128	1,024
Physical interface	Passive copper: NVLink cable-cartridge / backplane	Electrical/copper	Electrical/copper	Hybrid: orthogonal cableless electrical interconnect intra-rack; optical interconnect inter-rack	All-copper Cable Tray	Electrical/copper	Cableless OEX electrical interconnect	Backplane-free orthogonal electrical; NPC interconnect	Cableless OEX electrical interconnect	NPO optics	SiPh optical circuit switching supplied by Lightelligence	NPO optics
Copper cable	Yes	Yes	Yes	No (cableless)	Yes	Undisclosed	No (cableless)	Yes	No (cableless)	No	No	No
CPO	No for scale-up (Yes for scale-out)	No	No	Undisclosed	No	Undisclosed	No	No	No	No	No	No
NPO	No	No	No	Undisclosed	No	Undisclosed	No	No	No	Yes	No	Yes

Source: Company data, Morgan Stanley Research

Exhibit 5: Scale-up vs. scale-out networking



Source: Broadcom 2025 OCP Summit Keynote, Morgan Stanley Research

China AI server scale-up development

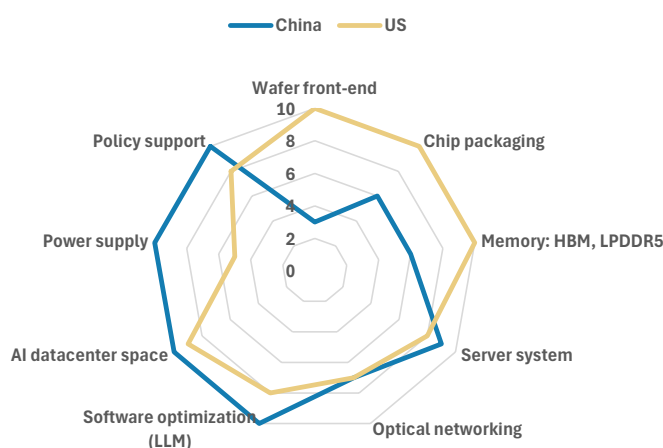
In China, electrical interconnect remains the preferred option within racks, while optical becomes more important as scale-up domains extend across racks. Short electrical links offer lower cost and simpler integration within boards, trays and single-rack systems. Moore Threads' C256 uses an all-copper cable tray design, while the orthogonal zero-cable architectures by Enflame and MetaX also remain electrical. Optical interconnection becomes more attractive across racks, where copper reach, signal loss and power consumption become constraints. Huawei's Atlas 950 SuperPoD extends UnifiedBus across racks within the same SuperPoD using optical interconnect.

The transition toward optical connectivity may create demand for domestic switching and optical engine solutions designed for AI systems. At WAIC 2026, Lightelligence (not covered) showcased its new NPO/CPO switch solutions (Exhibit 8, Exhibit 9), which pair the switch ASIC from a leading domestic vendor, which we believe is Centec Communications (not covered), with Lightelligence's optical engine, including the SiPh PIC and selected EICs such as the TIA. The NPO/CPO solutions shorten the high-speed

electrical path between the switch ASIC and optical engine, improving bandwidth density and power efficiency, while NPO retains greater serviceability than CPO. The same architecture can serve both scale-up and scale-out networks, with scale-up becoming increasingly relevant as tightly coupled accelerator domains extend across racks.

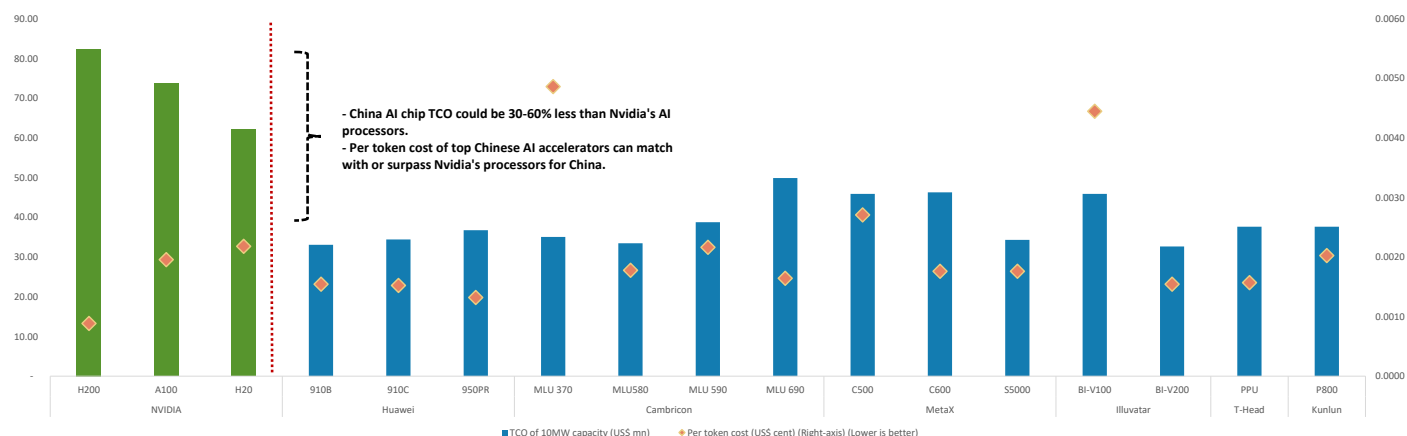
More broadly, scale-up is emerging as the principal area of architectural differentiation, while scale-out remains the backbone of large-scale clusters. System solutions generally use a specialized fabric to connect accelerators within a supernode, and Ethernet-based fabrics such as RoCE or UBoE, or Infiniband to connect multiple supernodes across a cluster. NVIDIA combines NVLink with InfiniBand or Spectrum-X Ethernet; AMD Helios pairs UALink with Ethernet scale-out; Huawei uses UnifiedBus within the SuperPoD, and UBoE or RoCE for scale-out between SuperPoDs.

Exhibit 6: Relative strengths of AI industries in the US and China - optical networking almost on par



Source: Morgan Stanley Research

Exhibit 7: Domestic chips have lower TCO and comparable per token cost (AI LLM inference) vs. NVIDIA's processors for China

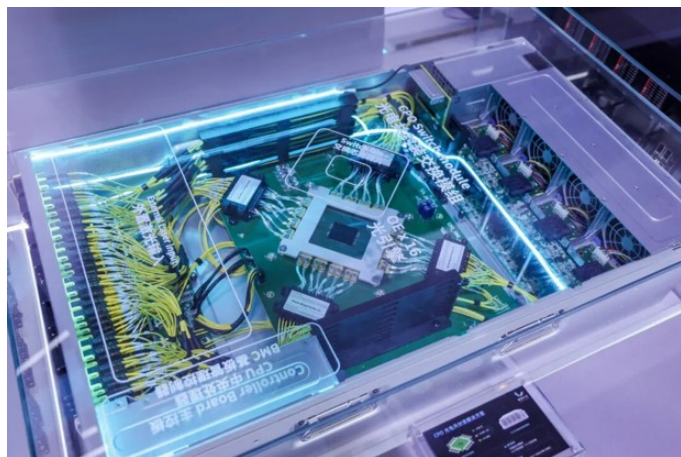


Source: Company data, Morgan Stanley Research estimates

Optical interconnect is progressing towards scale-up system deployment

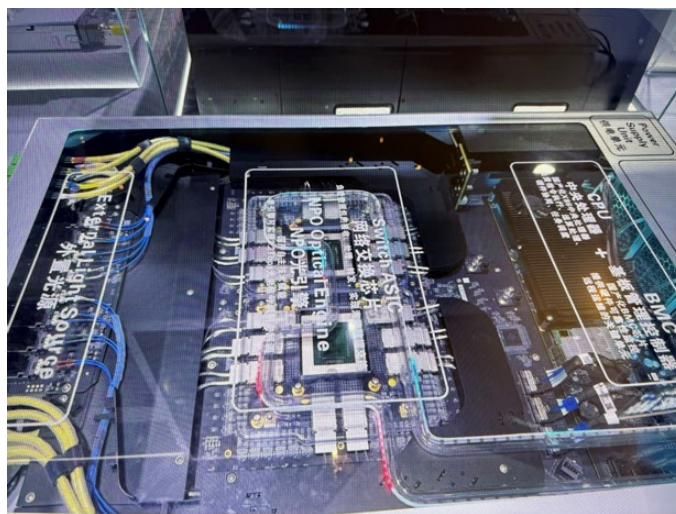
Enclave and Lightelligence previously demonstrated an xPU-CPO prototype that places the optical engine beside the accelerator, enabling direct optical output from the xPU and reducing the electrical distance between the compute silicon and optics. At WAIC 2026, Enclave displayed an NPO-equipped GPU-server solution targeting scale-up domains of 512 accelerators or more, developed in partnership with Lightelligence ([Exhibit 10](#)). Biren also unveiled a next-gen BR2xx-based NPO architecture designed to support a scale-up domain of up to 1,024 GPUs ([Exhibit 11](#)). Also, LightSphere X (jointly developed by Lightelligence, Biren, ZTE, and other partners) incorporates Lightelligence's SiPh interconnect and dOCS technology, enabling the accelerator domain to extend across racks and allowing the topology to be reconfigured according to workload requirements ([Exhibit 12](#)). These projects show domestic optical technology being integrated at both the accelerator optical-I/O layer and the distributed optical-switching layer, to support larger, multi-rack scale-up domains.

Exhibit 8: 51.2T CPO switch module (displayed by Lightelligence)



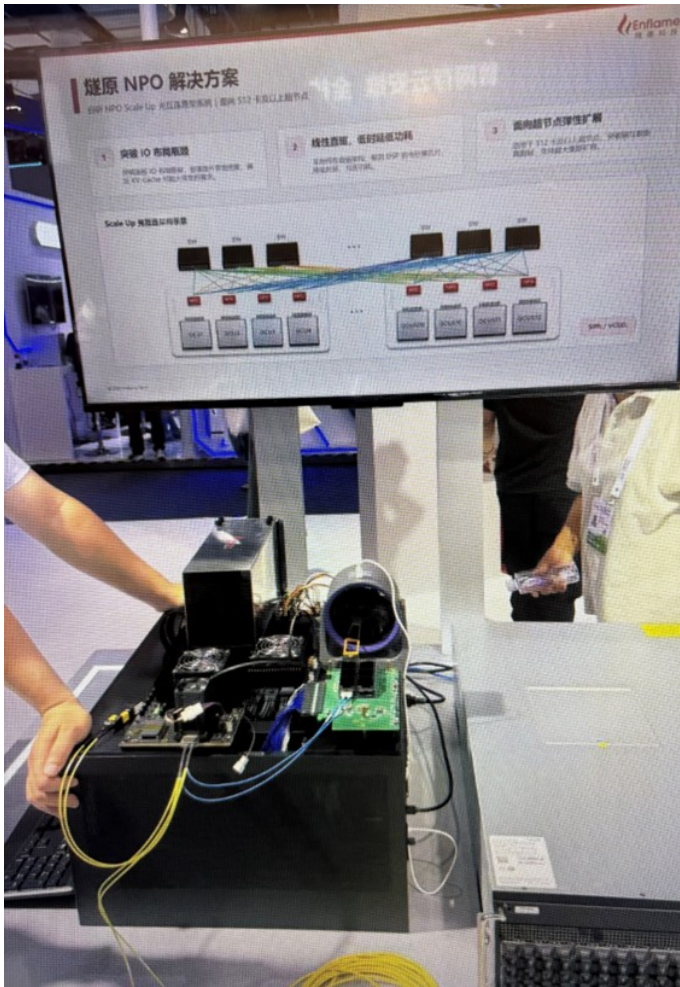
Source: Company data, Morgan Stanley Research

Exhibit 9: NPO switch module (displayed by Lightelligence)



Source: Company data, Morgan Stanley Research

Exhibit 10: Enflake's NPO optical interconnect solution for scale-up



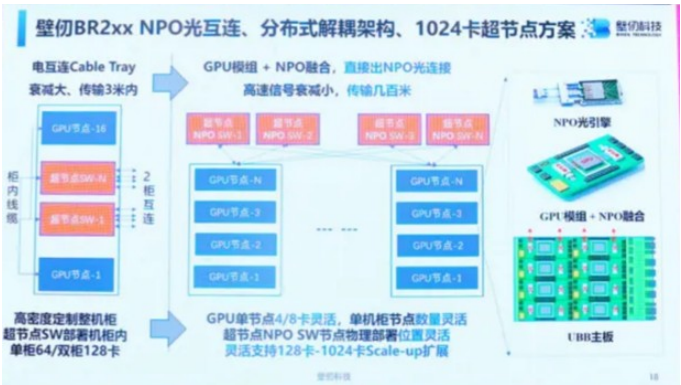
Source: Company data

Exhibit 12: LightSphere X (Lightelligence)



Source: Company data, Morgan Stanley Research

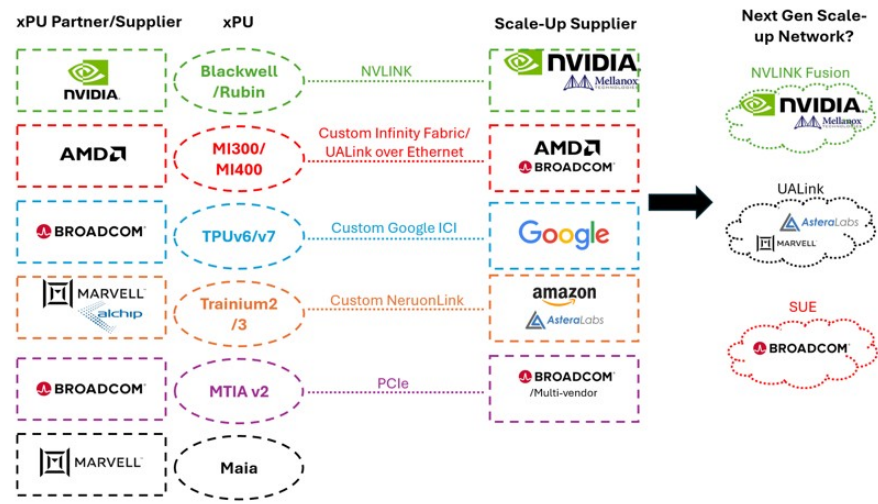
Exhibit 11: Biren's NPO optical interconnect solution for scale-up



Source: Company data

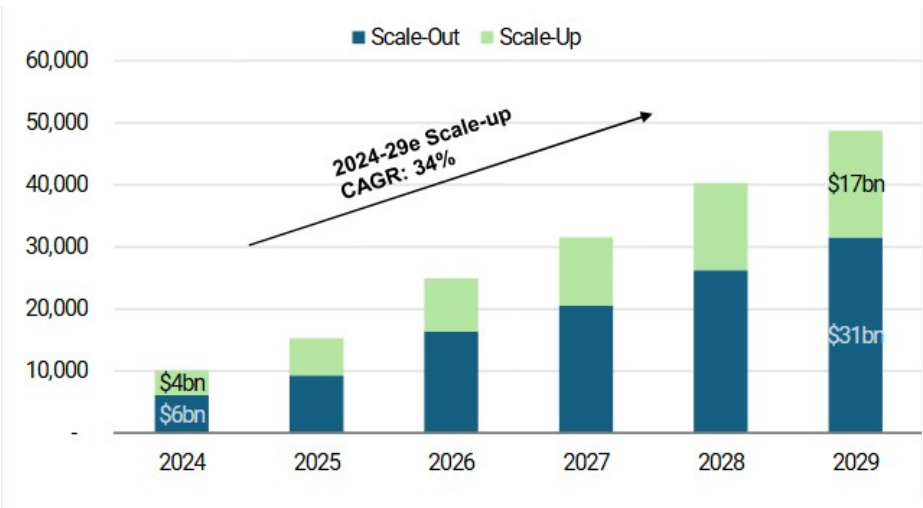
US scale-up networking ecosystem and TAM forecast

Exhibit 13: Chinese CSPs' capex will be a key demand driver for Chinese AI GPUs



Source: Company data, Morgan Stanley Research

Exhibit 14: US semi : Scale-up TAM estimate (\$mn)



Source: Company data, Morgan Stanley Research (E) estimates

This report references U.S. Executive Order 14032 and/or entities or securities that are designated thereunder. U.S. persons may be prohibited from buying certain securities of entities named in this report. Readers are solely responsible for ensuring that their investment activities are carried out in compliance with applicable laws.

This report references export controls and/or entities that may be subject to export control restrictions. Readers are solely responsible for ensuring that their investment or trade activities are carried out in compliance with applicable laws.

This report references U.S. Executive Order 14105 and/or entities that may be in scope of such order. U.S. persons may be prohibited from engaging in certain transactions or otherwise require certain other transactions be notified to the U.S. Department of Treasury. Readers are solely responsible for ensuring that their investment or trade activities are carried out in compliance with applicable laws.

Valuation Methodology and Risks

Montage Technology Co Ltd (6809.HK)

Base case, residual income model. Key assumptions:

- 8.4% CoE (1.2 beta, 3.0% risk-free rate, and 4.5% risk premium)
- 30% payout ratio
- 19.3% medium-term growth rate
- 4% terminal growth rate

These assumptions reflect cloud capex growth, DRAM interface technology migration, and Montage's strong position in China's datacenter semi localization. We then apply an exchange rate of 1.15 HKD:1 RMB, assuming no H-share discount vs the A-share.

Risks to Upside

- Faster-than-expected phase-out of US peers
- Faster-than-expected spec migration

Risks to Downside

- Weaker-than-expected cloud demand
- Slower-than-expected DRAM interface technology migration
- Delay in new product launches

Montage Technology Co Ltd (688008.SS)

Base case, residual income model. Key assumptions:

- 8.4% CoE (1.2 beta, 3.0% risk-free rate, and 4.5% risk premium)
- 30% payout ratio
- 19.3% medium-term growth rate
- 4% terminal growth rate

We believe these assumptions are justified, given the cloud capex growth, DRAM interface technology migration, and Montage's strong position in China's datacenter semi localization.

Risks to Upside

- Faster-than-expected phase-out of US peers
- Faster-than-expected spec migration

Risks to Downside

- Weaker-than-expected cloud demand
- Slower-than-expected DRAM interface technology migration
- Delay in new product launches

Hygon Information Technology Co., Ltd. (688041.SS)

We assume an 8.0% cost of equity (beta 1.05, risk-free rate 2.0% and risk premium 5.8%), a payout ratio of 50%, a medium-term growth rate of 25.0%, and a terminal growth rate of 5.0%, all of which are in line with other China AI chip companies under our coverage.

Risks to Upside

- Hygon further differentiates itself from local CPU and GPU competitors and gains significant market share in China's CPU and GPU markets
- China AI demand is stronger than expected
- Faster ramp-up of leading node capacity

Risks to Downside

- Intensified pricing competition among local GPU companies
- China AI demand is weaker than expected
- Slower yield improvement and capacity buildout at local leading node foundries

Cambricon Technology Corporation (688256.SS)

Key valuation assumptions underpinning our model include: an 8.4% cost of equity (derived from a beta of 1.06, risk-free rate of 2.0%, and equity risk premium of 6.0%), a long-term payout ratio of 40% (was 57%), a medium-term growth rate of 16%, and a terminal growth rate of 6.0%.

Risks to Upside

- Stronger-than-expected AI demand
- CSP order ramp-up
- Accelerating localization

Risks to Downside

- Capacity and yield constraints
- Customer concentration risk
- Slower technology iteration

Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)

Key valuation assumptions include:

- An 8.3% cost of equity (derived from a beta of 1.05, risk-free rate of 2.0%, and equity risk premium of 6.0%)
- A long-term payout ratio of 34% (was 33%)
- A medium-term growth rate of 16%
- A perpetual terminal growth rate of 6%

Risks to Upside

- Stronger-than-expected CSP orders
- Faster CUDA replacement with Iluvatar's software
- Expansion of overseas and domestic capacity

Risks to Downside

- Order ramp below expectations
- Escalation of sanctions
- Intensifying competition

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley Asia Limited (which accepts the responsibility for its contents) and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research), and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Charlie Chan; Daisy Dai, CFA; Tiffany Yeh; Daniel Yen, CFA.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of June 30, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: ACM Research Inc, Advanced Micro-Fabrication Equipment Inc, Advanced Wireless Semiconductor Co, Alchip Technologies Ltd, AllRing Tech Co., AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMP Ltd, Cambricon Technology Corporation, Dosilicon Co Ltd, FOCI Fiber Optic Communications Inc, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Hangzhou Silan Microelectronics Co. Ltd., King Yuan Electronics Co Ltd, Macronix International Co Ltd, MediaTek, Montage Technology Co Ltd, Parade Technologies Ltd, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, SG Micro Corp., Shanghai Fudan Microelectronics, Silergy Corp., Silicon Motion, TSMC, UMC, Unigroup Guoxin Microelectronics Co Ltd, Vanguard International Semiconductor, WIN Semiconductors Corp, Winbond Electronics Corp, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Alchip Technologies Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Montage Technology Co Ltd, Powerchip Semiconductor Manufacturing Co.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from ASMP Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Montage Technology Co Ltd, Powerchip Semiconductor Manufacturing Co.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Advanced Micro-Fabrication Equipment Inc, Alchip Technologies Ltd, AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMP Ltd, Aspeed Technology, Espressif Systems, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Himax Technologies Inc, Hua Hong Semiconductor Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Innoscience, King Yuan Electronics Co Ltd, Macronix International Co Ltd, MediaTek, MetaX Integrated Circuits, Montage Technology Co Ltd, Novatek, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Realtek Semiconductor, SG Micro Corp., Shenzhen Longsys Electronics Co Ltd, SICC Co Ltd, Silergy Corp., Silicon Motion, TSMC, UMC, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, Winbond Electronics Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from ASE Technology Holding Co. Ltd., King Yuan Electronics Co Ltd, MediaTek, Nanya Technology Corp., Novatek, Nuvoton Technology Corporation, Realtek Semiconductor, Silicon Motion, SMIC, TSMC, UMC, Universal Scientific Ind. (Shanghai), Winbond Electronics Corp, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Advanced Micro-Fabrication Equipment Inc, Alchip Technologies Ltd, AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMP Ltd, Aspeed Technology, Espressif Systems, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Himax Technologies Inc, Hua Hong Semiconductor Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Innoscience, King Yuan Electronics Co Ltd, Macronix International Co Ltd, MediaTek, MetaX Integrated Circuits, Montage Technology Co Ltd, Novatek, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Realtek Semiconductor, SG Micro Corp., Shenzhen Longsys Electronics Co Ltd, SICC Co Ltd, Silergy Corp., Silicon Motion, TSMC, UMC, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, Winbond Electronics Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: ASE Technology Holding Co. Ltd., Iluvatar CoreX Semiconductor Co., Ltd., King Yuan Electronics Co Ltd, MediaTek, Montage Technology Co Ltd, Nanya Technology Corp., Novatek, Nuvoton Technology Corporation, Realtek Semiconductor, Silicon Motion, SMIC, TSMC, UMC, Universal Scientific Ind. (Shanghai), Winbond Electronics Corp, WT Microelectronics Co. Ltd..

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not

linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of June 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Cambricon Technology Corporation (688256.SS) - As of 07/27/26 GMT in CNY
Industry : Greater China Technology Semiconductors



Stock Rating History: 4/27/26 : O/A

Price Target History: 4/27/26 : 1065.77; 5/5/26 : 1342.28; 6/22/26 : 1528

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Hygon Information Technology Co., Ltd. (688041.SS) - As of 07/27/26 GMT in CNY
Industry : Greater China Technology Semiconductors



Stock Rating History: 7/3/26 : O/A

Price Target History: 7/3/26 : 480

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

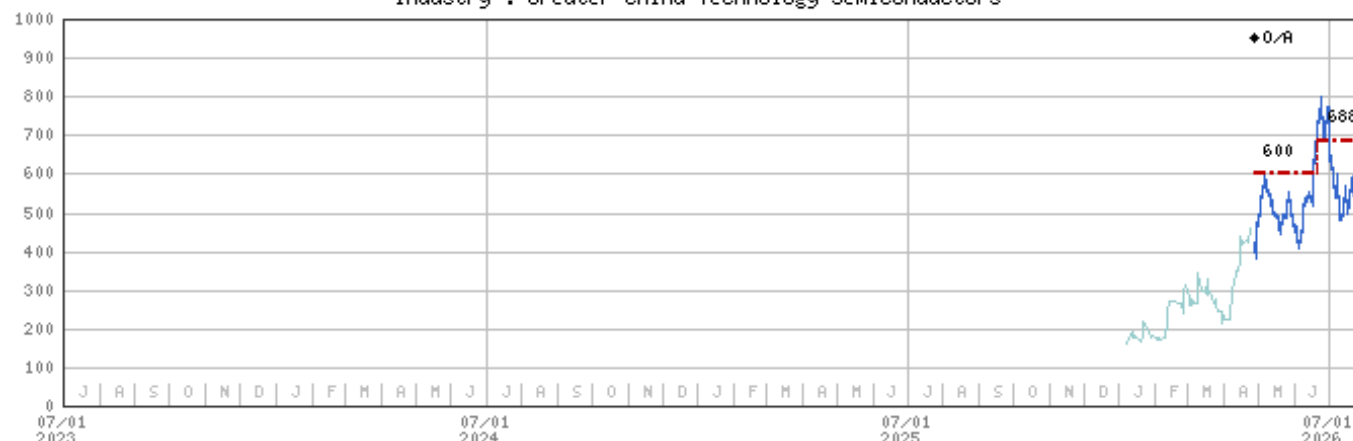
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK) - As of 07/27/26 GMT in HKD
Industry : Greater China Technology Semiconductors



Stock Rating History: 4/27/26 : O/A

Price Target History: 4/27/26 : 600; 6/22/26 : 688

Source: Morgan Stanley Research Date Format: MM/DD/YY Price Target: -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Montage Technology Co Ltd (6809.HK) - As of 07/27/26 GMT in HKD
Industry : Greater China Technology Semiconductors



Stock Rating History: 3/18/26 : O/A

Price Target History: 3/18/26 : 212; 4/28/26 : 310; 7/16/26 : 432

Source: Morgan Stanley Research Date Format: MM/DD/YY Price Target: -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

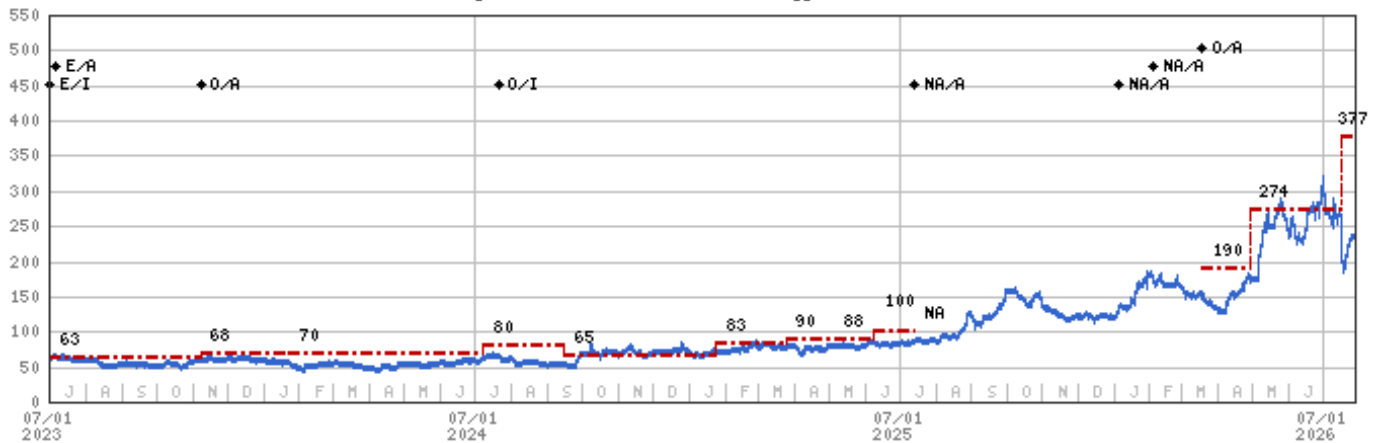
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Montage Technology Co Ltd (688008.SS) - As of 07/27/26 GMT in CNY
Industry : Greater China Technology Semiconductors



Stock Rating History: 7/1/21 : E/I; 10/12/21 : E/C; 5/20/22 : U/C; 10/4/22 : U/A; 12/13/22 : O/A; 2/22/23 : O/I; 5/4/23 : E/I; 7/7/23 : E/A; 11/8/23 : O/A; 7/21/24 : O/I; 7/14/25 : NA/A; 1/5/26 : NA/A; 2/4/26 : NA/A; 3/18/26 : O/A

Price Target History: 5/5/21 : 55; 10/29/21 : 62; 11/16/21 : 80; 4/8/22 : 60; 5/20/22 : 52; 7/19/22 : 44; 10/12/22 : 38; 12/13/22 : 85; 1/11/23 : 78; 5/4/23 : 63; 11/8/23 : 68; 1/23/24 : 70; 7/8/24 : 80; 9/15/24 : 65; 1/24/25 : 83; 3/25/25 : 90; 5/6/25 : 88; 6/9/25 : 100; 7/14/25 : NA; 3/18/26 : 190; 4/28/26 : 274; 7/16/26 : 377

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FCO118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 14-9169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital

Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of ASMT Ltd, GigaDevice Semiconductor Beijing Inc, Hua Hong Semiconductor Ltd, Montage Technology Co Ltd listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: <http://www.hkex.com.hk>.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/27/2026)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US\$84.32
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb399.77
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT\$121.50
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$3,460.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$608.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb1,241.02
Global Unichip Corp (3443.TW)	O (06/24/2026)	NT\$4,050.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT\$1,065.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT\$450.50
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK\$148.60
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK\$597.00
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$276.00
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb72.90
MediaTek (2454.TW)	O (11/28/2025)	NT\$3,680.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb830.20
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT\$436.00
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb770.00
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb95.31
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT\$1,820.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb98.89
Silergy Corp. (6415.TW)	U (05/19/2026)	NT\$449.00
SMIC (0981.HK)	O (10/21/2025)	HK\$70.65
TSMC (2330.TW)	O (02/07/2022)	NT\$2,350.00
UMC (2303.TW)	O (05/19/2026)	NT\$126.00
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT\$157.00
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT\$355.00

Daisy Dai, CFA

ASMP Ltd (0522.HK)	O (07/24/2025)	HK\$161.60
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb60.74
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb92.22
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb32.33
Hygon Information Technology Co., Ltd. (688041.SS)	O (07/03/2026)	Rmb313.59
Innoscence (2577.HK)	E (10/13/2025)	HK\$44.12
JCET Group Co Ltd (600584.SS)	E (01/16/2026)	Rmb82.35
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$24.88
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb100.00
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb84.88
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb66.00
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb24.77
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb93.22

Daniel Yen, CFA

AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT\$773.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT\$1,385.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT\$14,695.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT\$98.70
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT\$166.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb104.23
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb434.03
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT\$125.50
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK\$322.00
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb227.45
Novatek (3034.TW)	U (02/04/2026)	NT\$518.00
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT\$132.50
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT\$615.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT\$61.90
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT\$762.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb50.31
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT\$160.00
WPG Holdings (3702.TW)	O (03/16/2026)	NT\$116.00
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT\$203.50

Duan Liu

Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb115.15
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb373.90

Tiffany Yeh

AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT\$1,035.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT\$598.00
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US\$12.62
Hon Precision (7769.TW)	O (04/17/2026)	NT\$6,110.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT\$5,720.00
Silicon Motion (SIMO.O)	O (05/06/2024)	US\$270.90
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT\$6,395.00

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.